CMPG 322 - Decision Support Systems (DSS) 2
SU2 - Decision Trees
Introduction and example



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Outline

1. Development of accurate and useful decision trees
2. Bayesian Analysis
3. Utility Theory
4. Questions

Any problem that can be presented in a decision table can be graphically represented in a decision tree

- Most beneficial when a sequence of decisions must be made
- All decision trees contain decision points/nodes and state-of-nature points/nodes
- At decision nodes one of several alternatives may be chosen
- At state-of-nature nodes one state of nature will occur

Five steps of decision tree analysis

1. Define the problem
2. Structure or draw the decision tree
3. Assign probabilities to the states of nature
4. Estimate payoffs for each possible combination of alternatives and states of nature
5. Solve the problem by computing expected monetary values (EMVs) for each state of nature node

Structure of a Decision Tree

- Trees start from left to right
- Trees represent decisions and outcomes in sequential order
- Squares represent decision nodes
- Circles represent states of nature nodes
- Lines or branches connect the decisions nodes and the states of nature

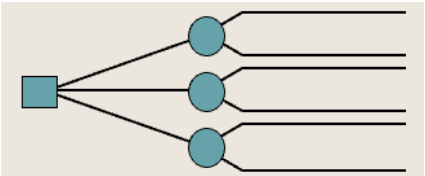
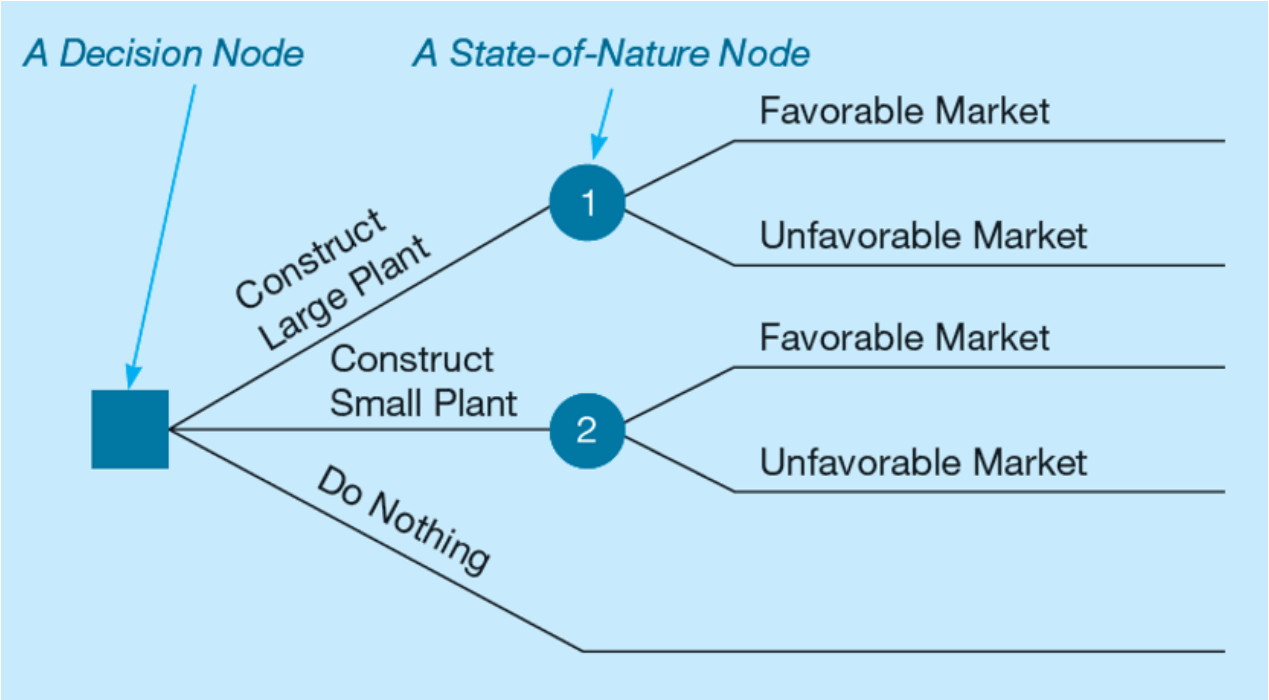


FIGURE 3.2 Thompson's Decision Tree



Example: Thompson Lumber Co.

FIGURE 3.2 Thompson's Decision Tree
Step 3: List the possible payoffs

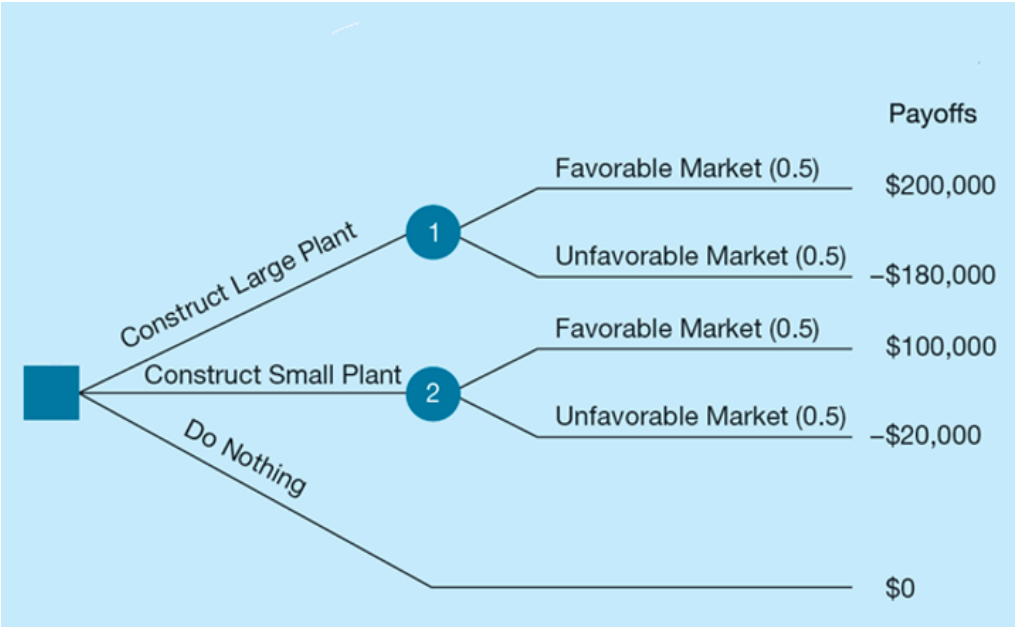


FIGURE 3.3 Completed and Solved Decision Tree for Thompson Lumber

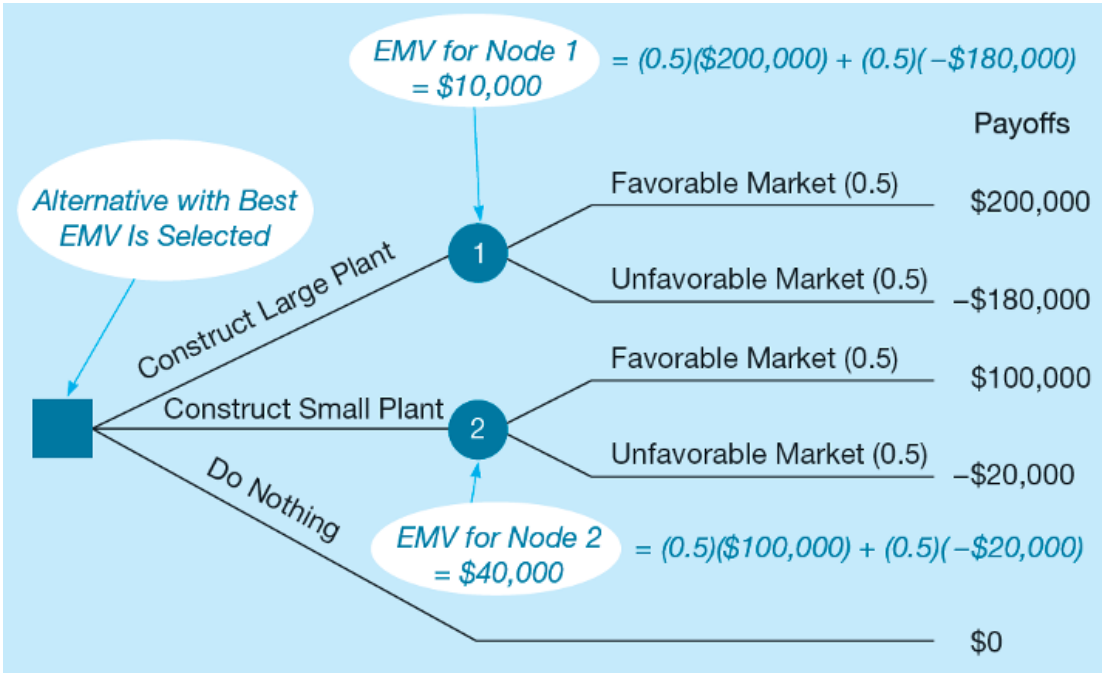
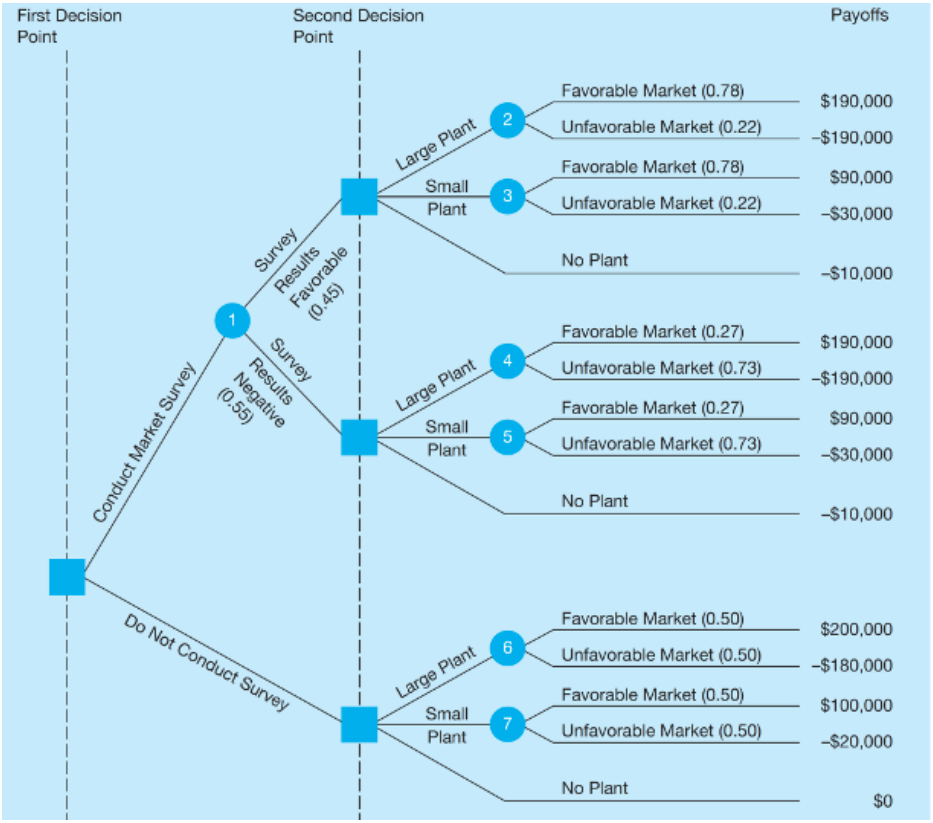


FIGURE 3.4 Larger Decision Tree with Payoffs and Probabilities for Thompson Lumber



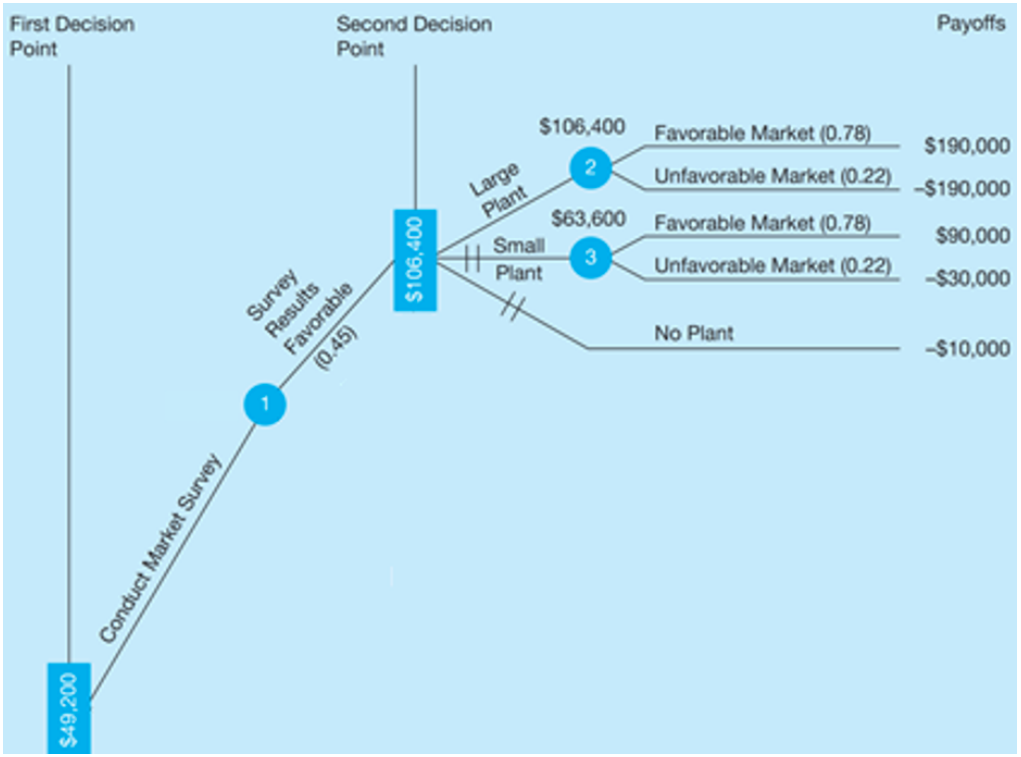
1. Given favorable survey results

$$EMV(node2) = EMV(large\ plant \mid positive\ survey)$$
$$= (0.78)(\$190,000) + (0.22)(-\$190,000)$$
$$= \$106,400$$

$$EMV(node3) = EMV(small\ plant \mid positive\ survey)$$
$$= (0.78)(\$90,000) + (0.22)(-\$30,000)$$
$$= \$63,000$$

$$EMV\ for\ no\ plant = -\$10,000$$

Thompson's complex decision tree



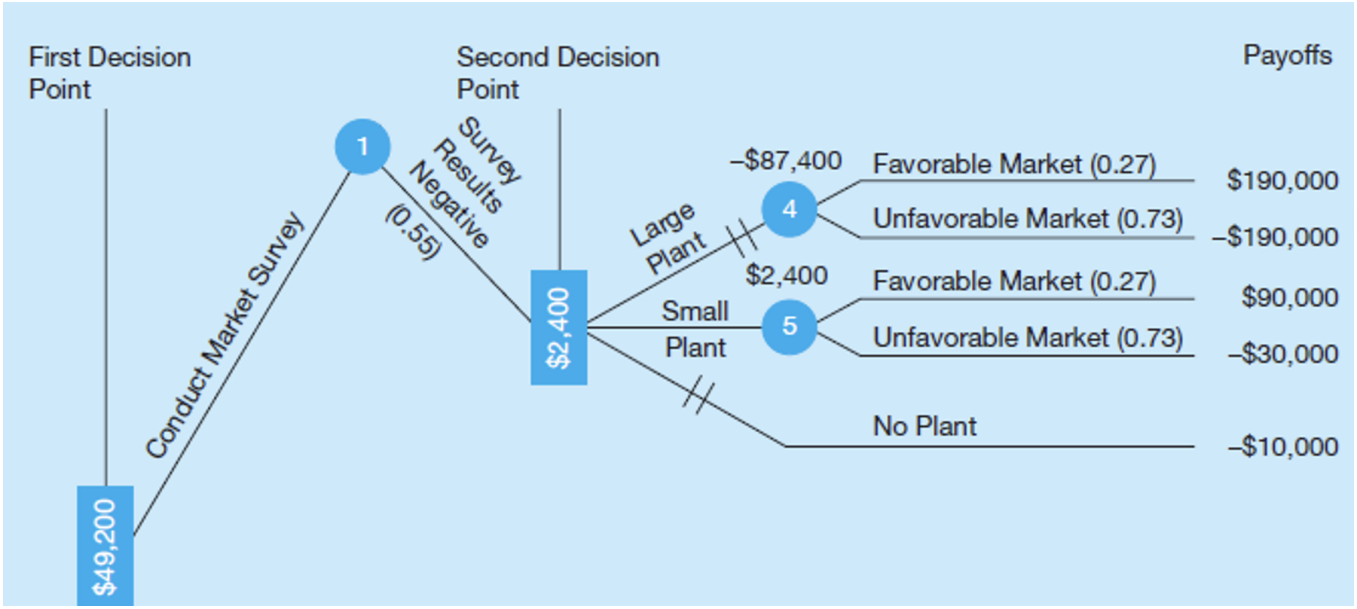
2 Given negative survey results

$$EMV(node4) = EMV(large\ plant \mid negative\ survey)$$
$$= (0.27)(\$190,000) + (0.73)(-\$190,000)$$
$$= -\$87,400$$

$$EMV(node5) = EMV(small\ plant \mid negative\ survey)$$
$$= (0.27)(\$90,000) + (0.73)(-\$30,000)$$
$$= \$2400$$

$$EMV\ for\ no\ plant = -\$10,000$$

Thompson's complex decision tree



3 Expected value of the market survey

$$EMV(node1) = EMV(conduct\ survey)$$
$$= (0.45)(\$106,400) + (0.55)(\$2,400)$$
$$= \$49,200$$

4 Expected value no market survey

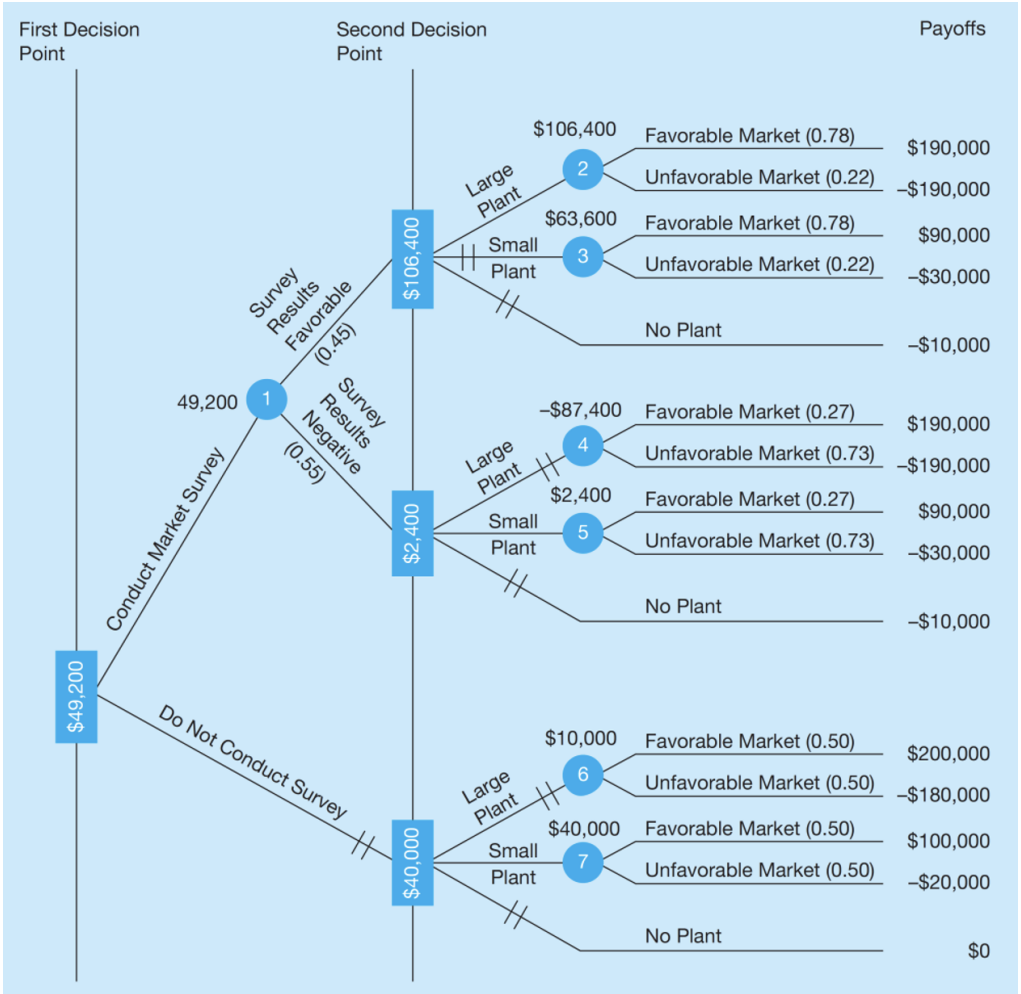
$$EMV(node6) = EMV(large\ plant)$$
$$= (0.50)(\$200,000) + (0.50)(-\$180,000)$$
$$= \$10,000$$

$$EMV(node7) = EMV(small\ plant)$$
$$= (0.50)(\$100,000) + (0.50)(-\$20,000)$$
$$= \$40,000$$

$$EMV\ for\ no\ plant\ = \$0$$

Thompson's complex decision tree

FIGURE 3.5 Thompson's Decision Tree with EMVs Shown



- Thompson wants to know the actual value of doing the survey:

$$EVSI = (Expected\ value\ with\ sample\ information) \times (Expected\ value\ of\ best\ decision\ without\ sample\ information) = (EV\ with\ SI\ +\ cost) - (EV\ without\ SI)$$

$$EVSI = (\$49,200 + \$10,000) - \$40,000 = \$19,200$$

Efficiency of sample information

- Possibly many types of sample information available
- Different sources can be evaluated

Expected Value of perfect Information

$$EVPI = EVwPI - Maximum\ EMV$$
$$= \$100,000 - \$40,000$$
$$= \$60,000$$

- How sensitive are the decisions to changes in the probabilities?
- How sensitive is our decision to the probability of a favorable survey result?
- If the probability of a favorable result ($p = 0.45$) were to change, would we make the same decision?
- How much could it change before we would make a different decision?

- p = probability of a favorable survey result
  $(1-p)$ = probability of a negative survey result
- EMV(node 1) = (\$106,400)p +(\$2,400)$(1-p)$
  = \$104,000p + \$2,400
- We are indifferent when the EMV of node 1 is the same as the EMV of not conducting the survey

\$104,000p + \$2,400 = \$40,000
\$104,000p = \$37,600
p = \$37,600÷\$104,000 = 0.36

If p < 0.36, do not conduct the survey
If p > 0.36, conduct the survey

- Many ways of getting probability data
  - Management's experience and intuition
  - Historical data
  - Computed from other data using Bayes' theorem
- Bayes' theorem incorporates initial estimates and information about the accuracy of the sources
- Allows the revision of initial estimates based on new information

Calculating Revised Probabilities

- Four known conditional probabilities for Thompson Lumber:
  P(favorable market(FM) | survey results positive) = 0.78
  P(unfavorable market(UM) | survey results positive) = 0.22
  P(favorable market(FM) | survey results negative) = 0.27
  P(unfavorable market(UM) | survey results negative) = 0.73
- Prior probabilities:
  P(FM) = 0.50
  P(UM) = 0.50

TABLE 3.16 Market Survey Reliability in Predicting States of Nature

	STATE OF NATURE	
RESULT OF SURVEY	FAVORABLE MARKET (FM)	UNFAVORABLE MARKET (UM)
Positive (predicts favorable market for product)	P (survey positive | FM) = 0.70	P (survey positive | UM = 0.20
Negative (predicts unfavorable market for product)	P (survey negative | FM) = 0.30	P (survey negative | UM) = 0.80

- Calculating posterior probabilities

$$P(A \mid B) = \frac{P(B \mid A) \times P(A)}{P(B \mid A) \times P(A) + P(B \mid A') \times P(A')}$$

Where:
A, B = any two events
A' = complement of A
A = favorable market
B = positive survey

Calculating Revised Probabilities

$$P(A \mid B) = \frac{P(B \mid A) \times P(A)}{P(B \mid A) \times P(A) + P(B \mid A') \times P(A')}$$

P(FM | survey positive)

$$= \frac{P(\text{survey positive} \mid \text{FM}) P(\text{FM})}{P(\text{survey positive} \mid \text{FM}) P(\text{FM}) + P(\text{survey positive} \mid \text{UM}) P(\text{UM})}$$

$$= \frac{(0.70)(0.50)}{(0.70)(0.50)+(0.20)(0.50)} = \frac{0.35}{0.45} = 0.78$$

P(UM | survey positive)

$$= \frac{P(\text{survey positive} \mid \text{UM}) P(\text{UM})}{P(\text{survey positive} \mid \text{UM}) P(\text{UM}) + P(\text{survey positive} \mid \text{FM}) P(\text{FM})}$$

$$= \frac{(0.20)(0.50)}{(0.20)(0.50)+(0.70)(0.50)} = \frac{0.10}{0.45} = 0.22$$

TABLE 3.17 Probability Revisions Given a Positive Survey

STATE OF NATURE	CONDITIONAL PROBABILITY *P*(SURVEY POSITIVE | STATE OF NATURE)	PRIOR PROBABILITY	JOINT PROBABILITY	POSTERIOR PROBABILITY *P*(STATE OF NATURE | SURVEY POSITIVE)
FM	0.70	× 0.50	= 0.35	0.35÷0.45 = 0.78
UM	0.20	× 0.50	= 0.10	0.10÷0.45 = 0.22
		P(survey results positive)	= 0.45	1.00

Calculating Revised Probabilities

PROGRAM 3.3 Results of Bayes' Calculations in Excel 2016

	A	B	C	D	E
1	**Bayes' Theorem for Thompson Lumber Example**				
2					
3	**Fill in cells B7, B8, and C7**				
4					
5	**Probability Revisions Given a Positive Survey**				
6	State of Nature	P(Sur.Pos	Prior Prob.	Joint Prob	Posterior Probability
7	FM	0.7	0.5	0.35	0.78
8	UM	0.2	0.5	0.1	0.22
9			P(Sur.pos.)=	**0.45**	
10					
11	**Probability Revisions Given a Negative Survey**				
12	State of Nature	P(Sur.Pos	Prior Prob.	Joint Prob	Posterior Probability
13	FM	0.3	0.5	0.15	0.27
14	UM	0.8	0.5	0.4	0.73
15			P(Sur.neg.)=	**0.55**	

	B	C	D	E
7	0.7	0.5	=B7*C7	=D7/$D$9
8	0.2	=1-C7	=B8*C8	=D8/$D$9
9		P(Sur.pos.)=	**=SUM(D7:D8)**	
10				
11				
12	P(Sur.Pos.|state of	Prior Prob.	Joint Prob.	Posterior Probability
13	=1-B7	=C7	=B13*C13	=D13/$D$15
14	=1-B8	=C8	=B14*C14	=D14/$D$15
15		P(Sur.neg.)=	**=SUM(D13:D14)**	

Calculating Revised Probabilities

$$P(A \mid B) = \frac{P(B \mid A) \times P(A)}{P(B \mid A) \times P(A) + P(B \mid A') \times P(A')}$$

P(FM | survey negative)

$$= \frac{P(\text{survey negative} \mid \text{FM}) P(\text{FM})}{P(\text{survey negative} \mid \text{FM}) P(\text{FM}) + P(\text{survey negative} \mid \text{UM}) P(\text{UM})}$$

$$= \frac{(0.30)(0.50)}{(0.30)(0.50)+(0.80)(0.50)} = \frac{0.15}{0.55} = 0.27$$

P(UM | survey negative)

$$= \frac{P(\text{survey negative} \mid \text{UM}) P(\text{UM})}{P(\text{survey negative} \mid \text{UM}) P(\text{UM}) + P(\text{survey negative} \mid \text{FM}) P(\text{FM})}$$

$$= \frac{(0.80)(0.50)}{(0.80)(0.50)+(0.30)(0.50)} = \frac{0.40}{0.55} = 0.73$$

Calculating Revised Probabilities

TABLE 3.18 Probability Revisions Given a Negative Survey

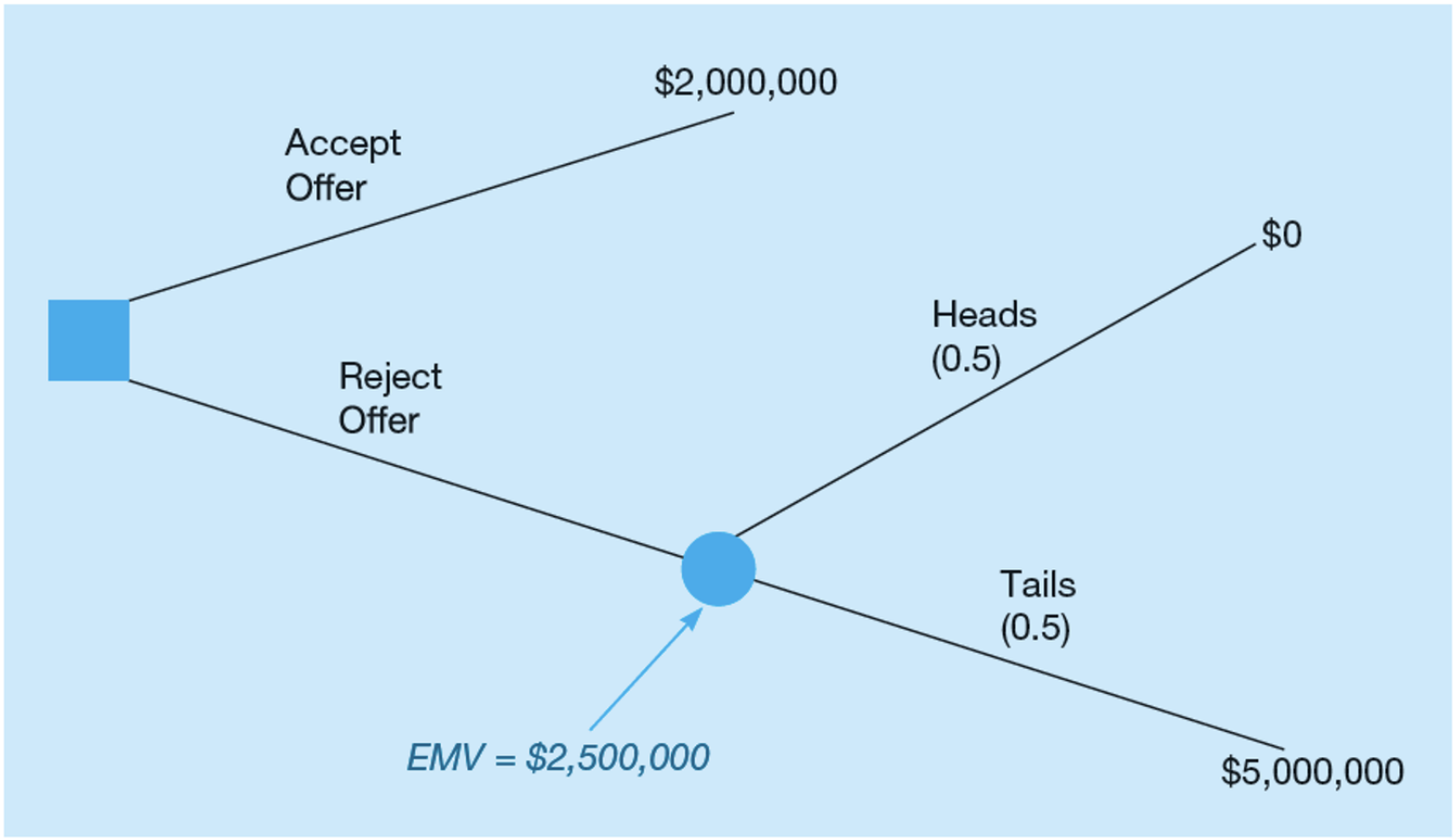
STATE OF NATURE	CONDITIONAL PROBABILITY *P*(SURVEY POSITIVE | STATE OF NATURE)	PRIOR PROBABILITY	JOINT PROBABILITY	POSTERIOR PROBABILITY *P*(STATE OF NATURE | SURVEY POSITIVE)
FM	0.30	× 0.50	= 0.15	0.15÷0.55 = 0.27
UM	0.80	× 0.50	= 0.40	0.40÷0.55 = 0.73
		P(survey results negative)	= 0.55	1.00

- Monetary value is not always a true indicator of the overall value of the result of a decision
- The overall value of a decision is called utility
- Economists assume that rational people make decisions to maximize their utility

Example:

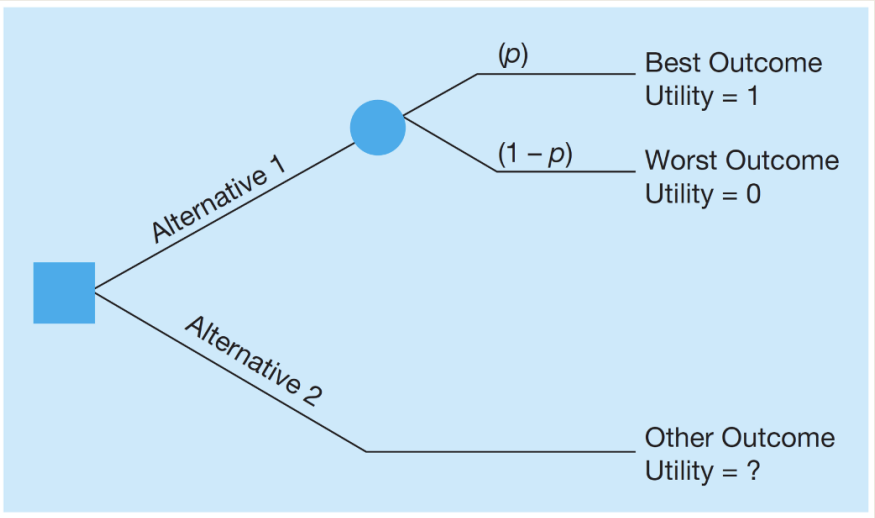
- You have a lottery ticket which could payout R5 million or nothing
- Someone offered you R2 million

FIGURE 3.6 Your Decision Tree for the Lottery Ticket



- Utility assessment assigns the worst outcome a utility of 0 and the best outcome a utility of 1
- A standard gamble is used to determine utility values
- When you are indifferent, your utility values are equal

FIGURE 3.7 Standard Gamble for Utility Assessment



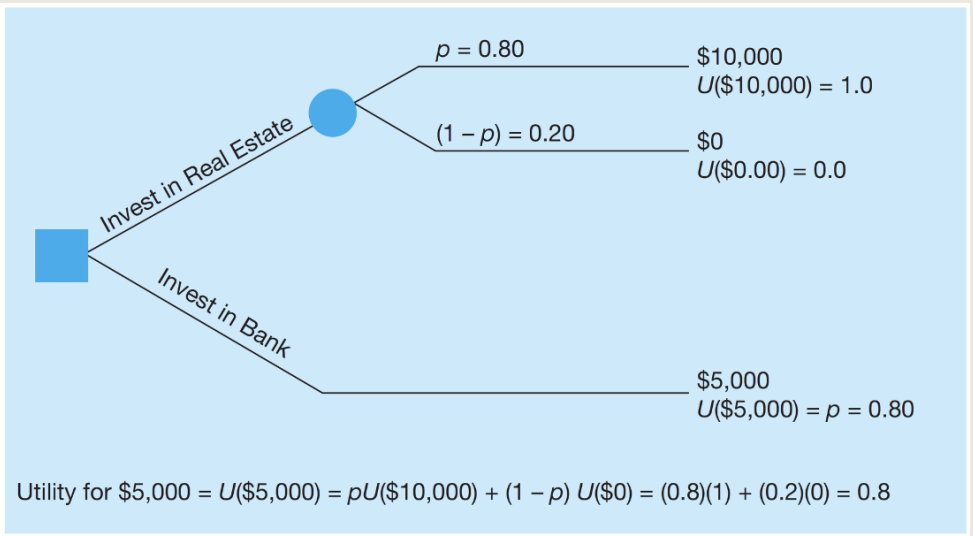
- Expected utility of alternative 2 = Expected utility of alternative 1
- Utility of other outcome = (p)(utility of *best* outcome, which is 1) + (1−p)(utility of the *worst* outcome, which is 0)
- Utility of other outcome = (p)(1) + (1−p)(0) = p

Constructing a utility curve

- Construct a utility curve revealing preference for money between \$0 and \$10,000
- A utility curve plots the utility value versus the monetary value
  - An investment in a bank will result in \$5,000
  - An investment in real estate will result in \$0 or \$10,000
  - Unless there is an 80% chance of getting \$10,000 from the real estate deal, prefer to have her money in the bank
  - If p = 0.80, Jane is indifferent between the bank or the real estate investment

Constructing a utility curve

FIGURE 3.8 Utility of $5,000



- Assess other utility values

  Utility for $7,000 = 0.90
  Utility for $3,000 = 0.50

- Use the three different dollar amounts and assess utilities

FIGURE 3.9 Utility Curve for Jan Dickson

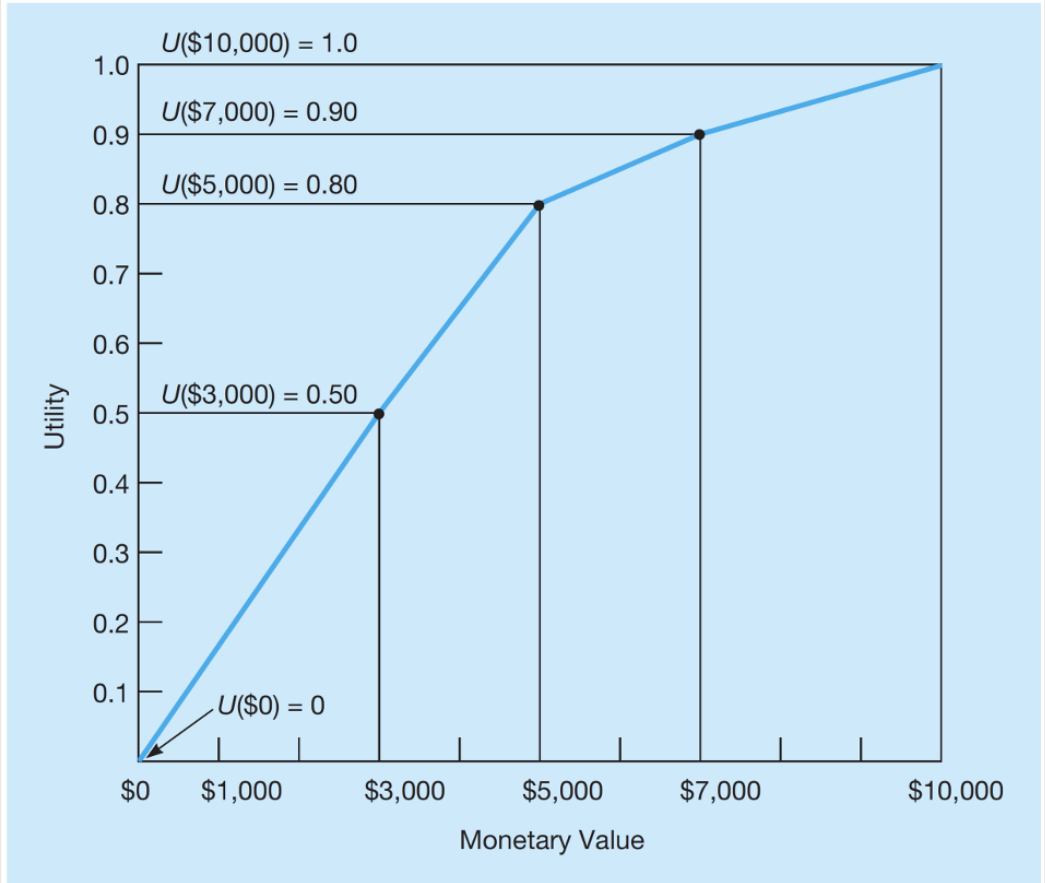
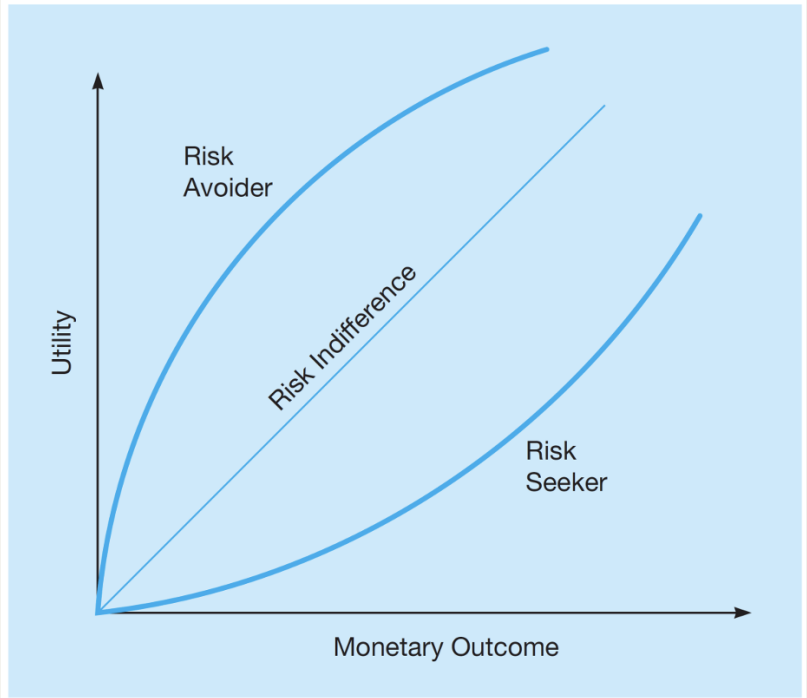


FIGURE 3.10 Preferences for Risk



- Typical of a **risk avoider**
  - *Less utility from greater risk*
  - *Avoids situations where high losses might occur*
  - *As monetary value increases, utility curve increases at a slower rate*
- A **risk seeker** gets more utility from greater risk
  - *As monetary value increases, the utility curve increases at a faster rate*
- Risk *indifferent* gives a linear utility curve

Utility as decision-making criterium

- Once a utility curve has been developed it can be used in making decisions
- Replaces monetary outcomes with utility values
- Expected utility is computed instead of the EMV

Questions

